

A Counterexample to Conjecture 00000008176: the Artin Density of a Perfect Square

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Claude (Opus 5.5)[†]

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Abstract

Let $\delta_A(q)$ be the density of the primes p modulo which q is a primitive root. The second claim of Conjecture 00000008176 is that $\delta_A(q)$ is zero or nonzero according to whether $q \not\equiv 1 \pmod{p^2}$ holds for every prime p . We show unconditionally that $\delta_A(s^2) = 0$ for every integer s . Since $q = 4$ satisfies the criterion and $q = 9$ violates it, the criterion decides nothing, in either direction. The refutation is checked in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000008176 reads:

Definition: The density table for computational verification of Artin's primitive root conjecture: the explicit values of the density $\text{delta_A}(q)$ of primes q being a primitive root (Hooley explicit). Conjecture: $\text{delta_A}(2) = 0.3739558136\dots$ (the explicit 2-value of the Artin constant, the product of $(1 - 1/(p(p-1)))$ with the Legendre correction); the complete classification of the values of $\text{delta_A}(q)$ (zero versus nonzero) is decided by q not congruent to 1 mod p^2 for all p (a non-Wieferich criterion); the minimal positive value of delta is attained at $q = 2$ (a 2-minimality law); and the arithmeticity of the density (rationality of delta): $\text{delta_A}(q)$ is rational if and only if q is a perfect square or -4 times a square (a rationality criterion), the list of rational values being finite (a finite rational list).

The Chinese version in `conjectures/00000008176.md` states the same four claims.

2 Reading of the statement

The conjecture is a conjunction of four claims. We refute the second one, so the conjecture is false whatever the other three claims mean.

The density. As in Artin's primitive root conjecture, $\delta_A(q)$ is the density of the set

$$\mathcal{A}(q) = \{p \text{ prime} : q \text{ is a primitive root modulo } p\}$$

relative to the primes. Here q is a primitive root modulo p if its residue generates the cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$, that is, it has multiplicative order $p-1$; a residue 0 is never a primitive root. The phrase “the density of primes q being a primitive root” can only mean this, since δ_A is indexed by q . The fourth claim speaks of q being a perfect square or -4 times a square, so q ranges over the integers, including perfect squares.

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Definition 1. A set S of primes has *density* d if $\#\{p \in S : p \leq x\}/\pi(x) \rightarrow d$ as $x \rightarrow \infty$, where $\pi(x)$ is the number of primes up to x . We write $\delta_A(q) = 0$ if $\mathcal{A}(q)$ has density 0.

The criterion. Let $W(q)$ be the statement “ $q \not\equiv 1 \pmod{p^2}$ for every prime p ”. The name “non-Wieferich criterion” indicates that p runs over primes. The second claim says that W decides whether $\delta_A(q)$ is zero or nonzero. The natural reading is

$$\delta_A(q) \neq 0 \iff W(q) \quad \text{for every integer } q. \quad (\text{N})$$

We also refute the reversed matching

$$\delta_A(q) = 0 \iff W(q) \quad \text{for every integer } q. \quad (\text{Z})$$

For (N) it does not matter whether “ $\delta_A(q) \neq 0$ ” is read as “ $\mathcal{A}(q)$ does not have density 0” or as “ $\mathcal{A}(q)$ has a density $d \neq 0$ ”.

3 Results

Proposition 2. For every integer s , $\mathcal{A}(s^2) \subseteq \{2\}$. Consequently $\delta_A(s^2) = 0$.

Theorem 3. $W(4)$ holds and $\delta_A(4) = 0$, so (N) fails. $W(9)$ fails and $\delta_A(9) = 0$, so (Z) fails. In particular Conjecture 00000008176 is false.

4 Proofs

Proof of Proposition 2. Let p be an odd prime and a the residue of s modulo p . If $a = 0$, the residue of s^2 is 0, which is not a primitive root. Otherwise Fermat’s little theorem gives

$$(a^2)^{(p-1)/2} = a^{p-1} = 1,$$

so the order of a^2 divides $(p-1)/2$. Since $0 < (p-1)/2 < p-1$, this order is not $p-1$, and s^2 is not a primitive root modulo p . Hence $\mathcal{A}(s^2) \subseteq \{2\}$.

It follows that $\#\{p \in \mathcal{A}(s^2) : p \leq x\} \leq 1$, so

$$0 \leq \frac{\#\{p \in \mathcal{A}(s^2) : p \leq x\}}{\pi(x)} \leq \frac{1}{\pi(x)} \rightarrow 0,$$

because $\pi(x) \rightarrow \infty$. So $\mathcal{A}(s^2)$ has density 0. □

Proof of Theorem 3. If $4 \equiv 1 \pmod{p^2}$ for a prime p , then $p^2 \mid 3$, which is impossible as $p^2 \geq 4$. So $W(4)$ holds, and $\delta_A(4) = 0$ by Proposition 2 with $s = 2$. This contradicts (N) at $q = 4$ in either reading of “ $\delta_A(4) \neq 0$ ”: the density is 0, and a density is unique since it is a limit.

On the other hand $9 \equiv 1 \pmod{2^2}$, so $W(9)$ fails, while $\delta_A(9) = 0$ by Proposition 2 with $s = 3$. This contradicts (Z) at $q = 9$. □

5 Formalization

The Lean 4 project in `lean/` (file `lean/Submission/Basic.lean`) formalizes the definitions above and proves both results against Mathlib.

- `artinSet q` is $\mathcal{A}(q)$: the primes p with `IsPrimitiveRoot ((q : ZMod p)) (p - 1)`, using Mathlib’s notion of a primitive root of unity, i.e. the residue has order exactly $p-1$.

- `HasPrimeDensity S d` states that $(S \cap [0, x]).\text{ncard} / \text{Nat.primeCounting } x$ tends to d .
- `NonWieferich q` is $W(q)$, with $q \equiv 1 \pmod{p^2}$ expressed by Mathlib’s `Int.ModEq`.
- `ClassificationNonzero` is $(\mathbb{N})$ with “ $\delta_A(q) \neq 0$ ” read as “not density 0”; `ClassificationNonzero’` is $(\mathbb{N})$ with “has a density $d \neq 0$ ”; `ClassificationZero` is $(\mathbb{Z})$.
- `artinSet_sq_subset` and `hasPrimeDensity_sq` are Proposition 2. They use Mathlib’s Fermat little theorem `ZMod.pow_card_sub_one_eq_one` and `Nat.tendsto_primeCounting`.
- `nonWieferich_four`, `not_nonWieferich_nine`, and the refutations `not_classificationNonzero`, `not_classificationNonzero’`, `not_classificationZero`.
- `conjecture_00000008176_false` states $\neg(P \wedge \text{ClassificationNonzero} \wedge Q \wedge R)$ for arbitrary propositions P, Q, R standing for the other three claims.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms: `#print axioms` reports exactly `propext`, `Classical.choice` and `Quot.sound` for the three refutations.

6 Remarks

Remark 4. The refutation is unconditional: it only uses that squares have density 0. The expected classification is different from the stated one. If $q = -1$ or q is a perfect square, then $\delta_A(q) = 0$ (the integer -1 is a primitive root only modulo 2 and 3). Otherwise, under the generalized Riemann hypothesis, Hooley [1] proved that $\delta_A(q)$ exists and is positive. The criterion W is unrelated to this dichotomy in both directions: 4 satisfies W with $\delta_A(4) = 0$, and $10 \equiv 1 \pmod{3^2}$ violates W although $\delta_A(10) > 0$ under the generalized Riemann hypothesis.

Remark 5. We do not address the other three claims. Under the generalized Riemann hypothesis, Hooley’s theorem gives $\delta_A(2)$ as Artin’s constant $0.3739558136\dots$. Unconditionally, it is not known for any single integer q that q is a primitive root modulo infinitely many primes.

References

- [1] C. Hooley, *On Artin’s conjecture*, J. Reine Angew. Math. 225 (1967), 209–220.